SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I P. HARINI (192521402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Harini

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	5
Experimental Design	30	6
Use of Tools	20	4
Analysis of Results	15	3
Documentation	10	3

Total Marks 21 **/30**

Signature of the Faculty

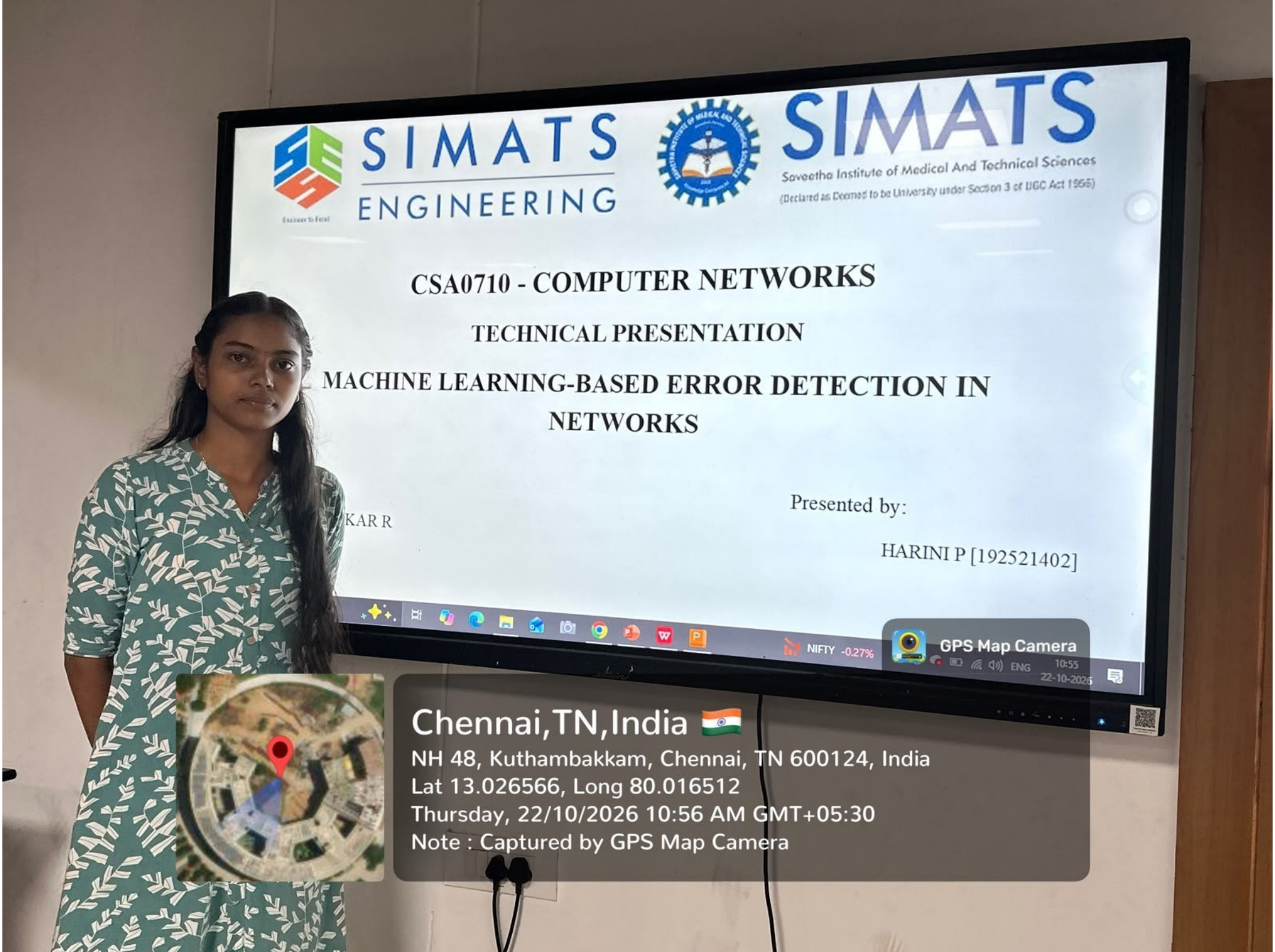
TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

			interpretation		
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation


SIMATS
ENGINEERING
SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)
CSA0710 - COMPUTER NETWORKS
TECHNICAL PRESENTATION
MACHINE LEARNING-BASED ERROR DETECTION IN NETWORKS
KAR R
Presented by:
HARINI P [192521402]
NIFTY -0.27%
GPS Map Camera
ENG
10:55
22-10-2026
Chennai,TN,India
NH 48, Kuthambakkam, Chennai, TN 600124, India
Lat 13.026566, Long 80.016512
Thursday, 22/10/2026 10:56 AM GMT+05:30
Note : Captured by GPS Map Camera

CSA0710 - COMPUTER NETWORKS

TECHNICAL PRESENTATION

MACHINE LEARNING-BASED ERROR DETECTION IN NETWORKS

Guide:

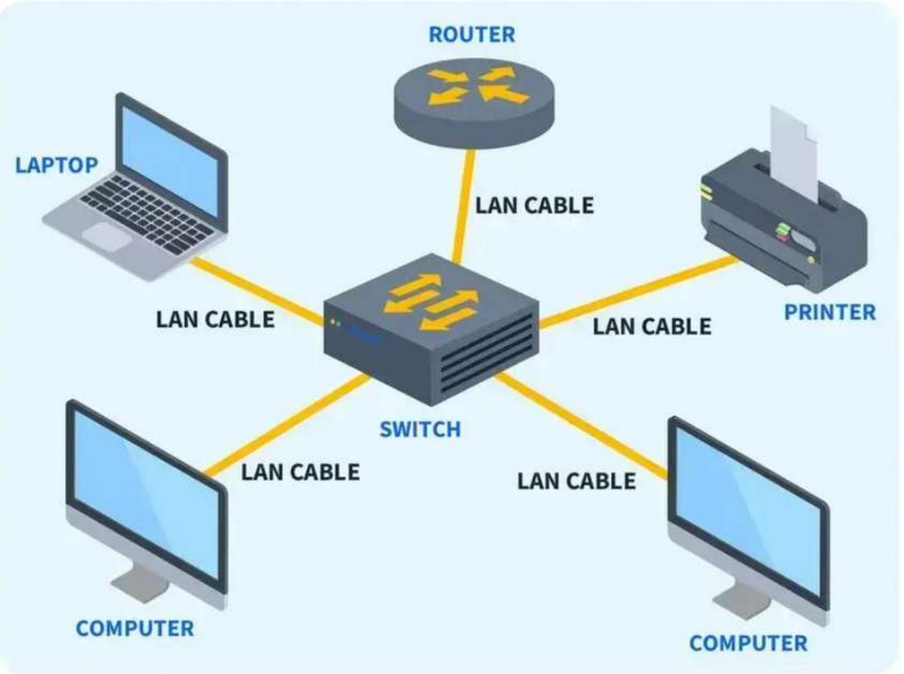
DR.SHANKAR R

Presented by:

HARINI P [192521402]

Introduction

- Computer networks enable communication between connected devices.
- Network errors affect data transmission and performance.
- Errors occur due to noise, congestion, and hardware failures.
- Traditional methods use predefined error detection algorithms.
- Machine Learning identifies patterns to detect network errors.

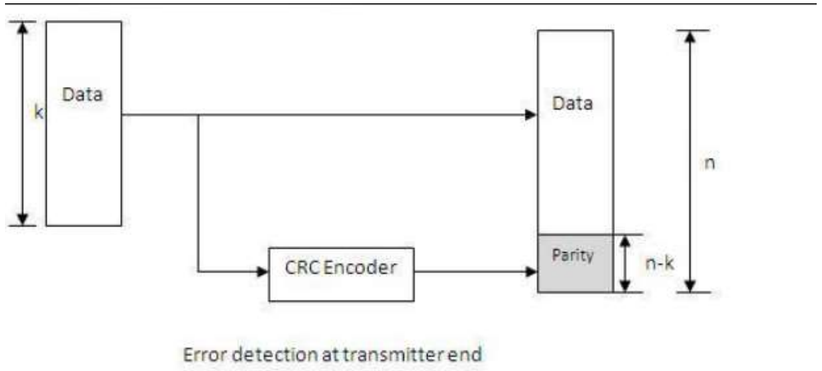


Types of Network Errors

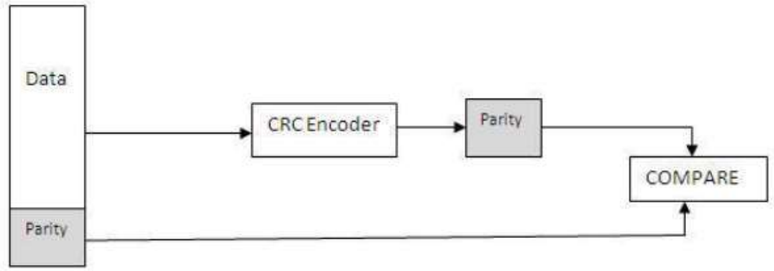
- Transmission Errors: Bits change during data transmission.
- Packet Loss: Packets fail to reach their destination.
- Network Congestion: Heavy traffic causes delays.
- High Latency: Data takes longer to reach its destination.
- Network Anomalies: Unusual traffic patterns indicate possible faults.

Traditional Error Detection Methods

- Parity Check: Detects certain bit errors using an extra bit.
- Checksum: Detects data changes using calculated values.
- CRC: Detects transmission errors using polynomial calculations.
- Rule-Based Monitoring: Identifies problems using fixed thresholds.
- Traditional methods have limited ability to learn complex patterns.



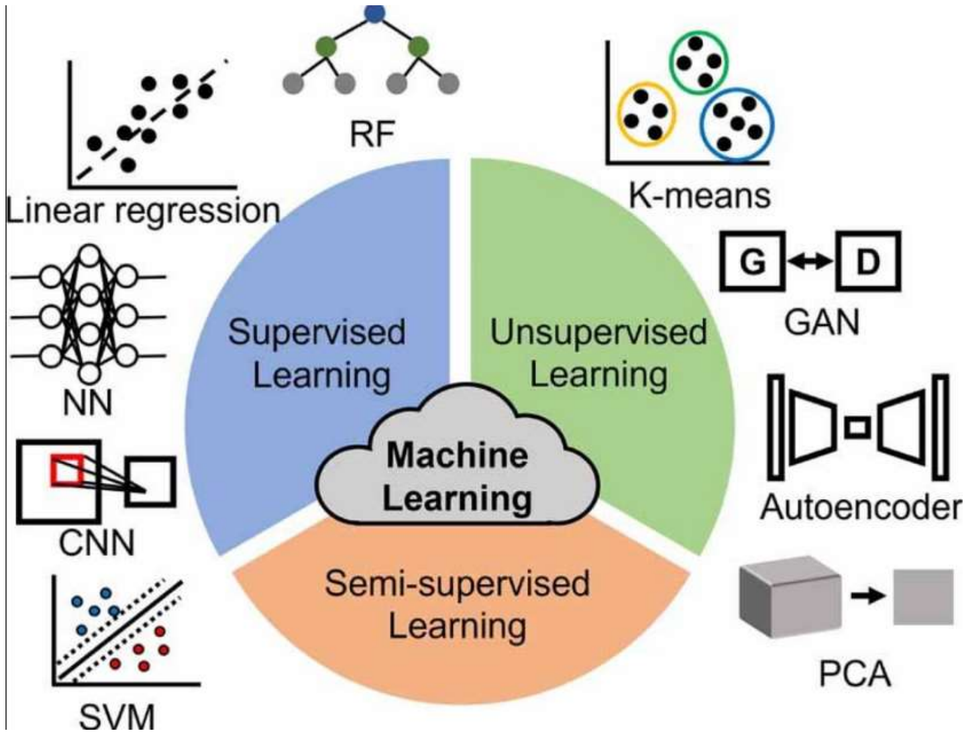
Error detection at transmitter end



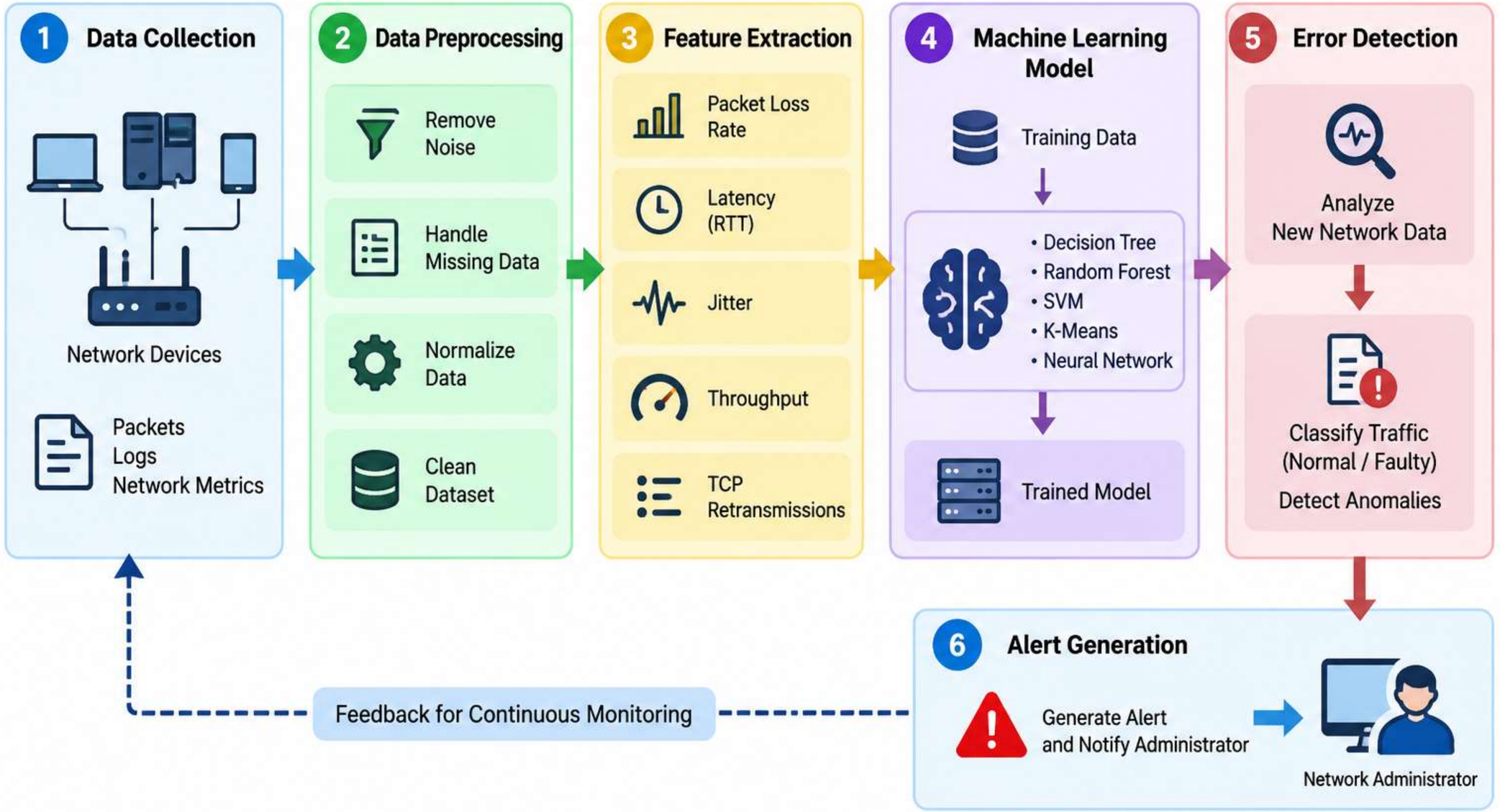
Error detection at receiver end

What Is Machine Learning?

- Machine Learning is a branch of Artificial Intelligence.
- It enables computers to learn patterns from data.
- Supervised Learning: Uses labeled network data.
- Unsupervised Learning: Identifies unusual patterns without labels.
- ML helps classify traffic and detect network anomalies.



Proposed System Architecture



Machine Learning Algorithms

- Decision Tree: Classifies traffic using decision rules.
- Random Forest: Combines multiple decision trees.
- SVM: Separates different network traffic classes.
- K-Means: Groups traffic and helps identify anomalies.
- Neural Networks: Learn complex patterns in network data.

Data Collection and Features

- Collect network traffic using monitoring tools.
- Measure packet loss and packet error counts.
- Record latency, jitter, and round-trip time.
- Monitor throughput and TCP retransmissions.
- Use Wireshark, Python, and datasets such as CIC-IDS2017.

Model Training and Detection

- Collect and label network traffic data.
- Clean the data and select relevant features.
- Split the dataset into training and testing sets.
- Train and evaluate the selected ML algorithm.
- Detect suspicious traffic patterns and generate alerts.

Example Scenario

- A college Wi-Fi network experiences frequent connection problems.
- The system collects packet loss and latency measurements.
- The ML model analyzes the collected network data.
- The model flags patterns associated with possible network faults.
- The administrator investigates congestion or hardware problems.

Advantages, Challenges and Applications

- Advantage: Detects complex network traffic patterns.
- Advantage: Supports automated network monitoring.
- Challenge: Requires good-quality training data.
- Challenge: May generate false alarms.
- Applications: Campus networks, cloud computing, IoT, and telecommunications.

Conclusion and Future Scope

- ML improves the detection of unusual network behavior.
- It complements traditional error detection techniques.
- Automated monitoring helps reduce troubleshooting time.
- Future systems can use deep learning and edge AI.
- Intelligent fault diagnosis can improve network reliability.


Thank
you